

Foundations of Programming- CSE10010

Mini Project Presentation 2025-26

Smart Parking Management System

Division 11

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An illustration of a parking lot from a top-down perspective. It shows several parking spaces with white lines. A blue car is parked in the top right space, with a white sensor icon next to it. Below it, a white car is parked in the bottom left space. To its right, two more blue cars are parked in adjacent spaces. A white arrow points upwards in the space between the white car and the first blue car. The background is a light blue gradient.

Introduction and Motivation

The Problem with Manual Parking Management

Traditional parking management systems are often inefficient, leading to inaccurate tracking, frequent billing disputes, and customer dissatisfaction. The absence of dynamic pricing capabilities results in missed revenue opportunities, while reliance on manual processes makes systems prone to human error and lacks transparency in fare calculation. These significant challenges underscore the urgent need for a more advanced, automated, and reliable parking solution.

Problem Statement and Objectives

Problem Statement

- Inefficient tracking of parking spaces
- Manual processes leading to errors and disputes
- Lack of dynamic pricing and revenue optimization
- Absence of real-time occupancy data
- Customer dissatisfaction due to parking difficulties

Objectives

- Automate vehicle detection and space management
 - Ensure accurate and transparent billing
 - Implement dynamic pricing for revenue maximization
 - Provide real-time parking availability updates
 - Enhance user experience and reduce congestion
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Individual Contribution

Name	Work (Report/Presentation/Algorithm Design/Implementation)
Chinmay Rane	Implementation and Presenting
Rohit Bhanose	Implementation, Design and Report
Shravasti Ganvir	Report and Presentation
Kushagra Raj	Implementation, Design and Report

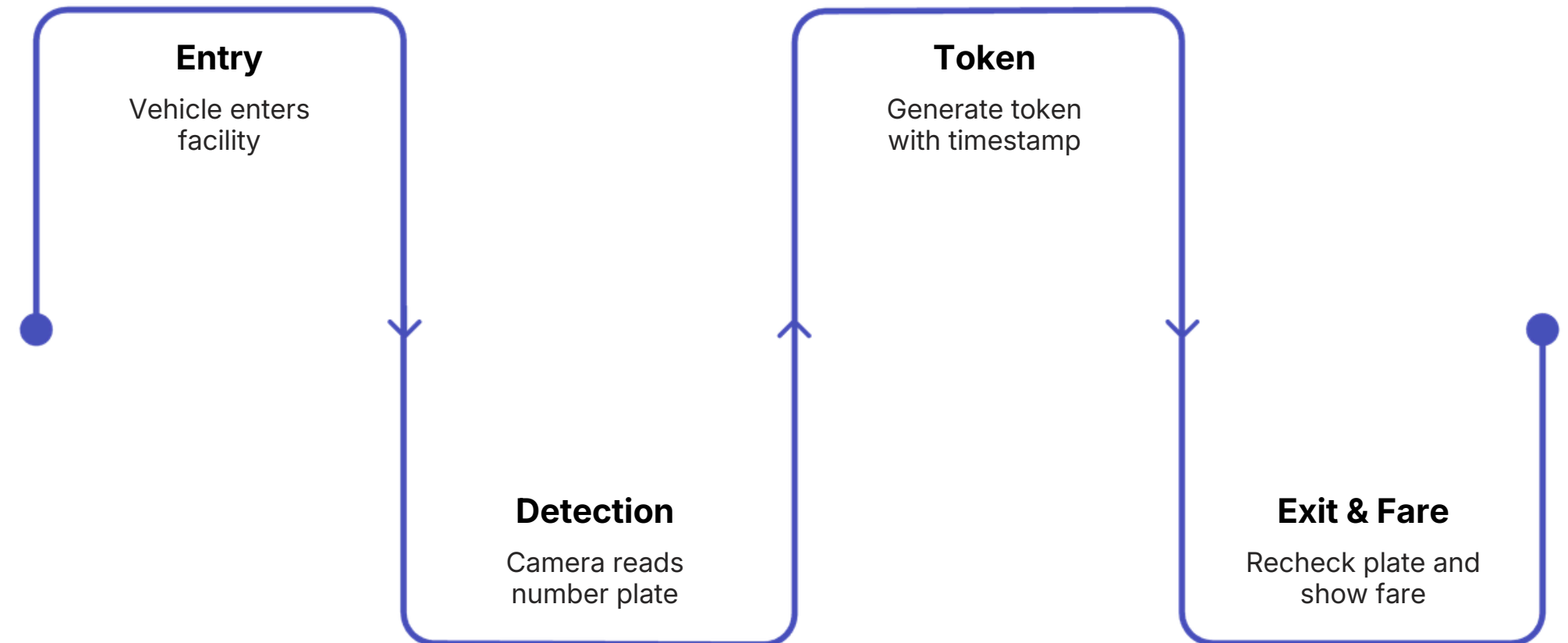
Proposed System

Detailed Description

Key features include:

- Camera-assisted vehicle identification using number plate recognition for seamless entry.
- A token-based workflow that precisely records entry timestamps, crucial for accurate billing.
- Real-time billing capabilities with dynamic fare calculation, adapting to various pricing models.
- Automatic detection of peak hours, allowing for the application of appropriate surcharges to optimize revenue and manage demand.

Flowchart/System Design



System Workflow



Entry

Vehicle enters the parking lot

Detection

Camera reads vehicle type and license plate



Token

Generate unique token with entry time

Exit & Fare

Plate rechecked and fare displayed

Dynamic Pricing Logic

Base Rates

Two-wheeler: ₹25 per hour

Four-wheeler: ₹40 per hour

Peak Hours

6:00 PM – 9:00 PM

Surcharge: +₹10 per hour during peak overlap

How It Works

Our pricing algorithm calculates total parking duration, identifies any overlap with peak hours, and applies appropriate rates.

- Base fare calculated for entire duration
- Peak hours identified automatically
- Surcharge applied only to peak overlap
- Transparent breakdown displayed to user

Implementation and Experimental Setup

Our solution, built for efficiency and reliability, was rigorously tested under various conditions.



C Language

Core logic for efficient performance.



Fully Automated

Minimal human interaction; machine-driven.



Redis-type Storage

Memory efficient; data stored temporarily.



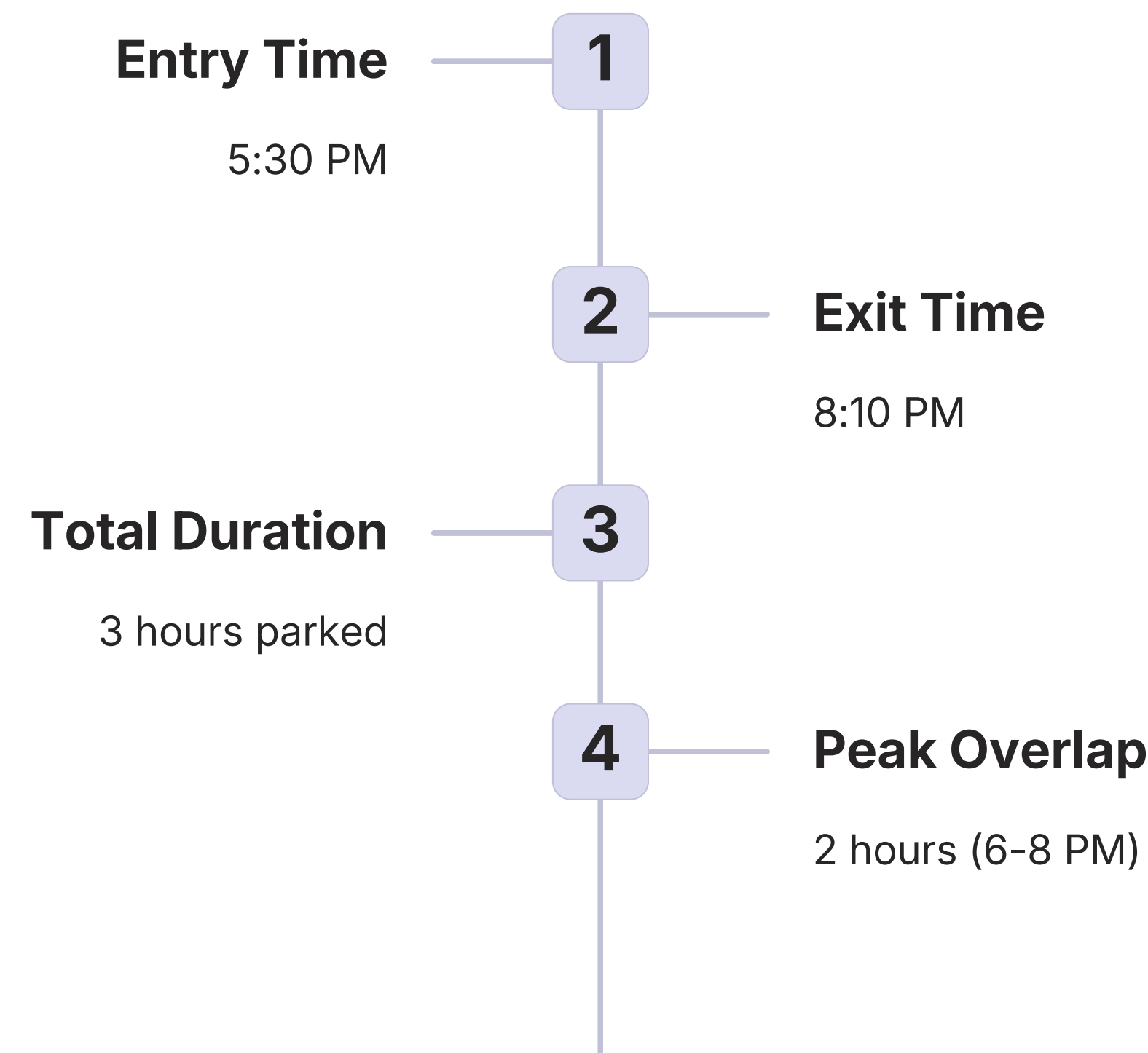
Simulated Camera

Simulates hardware camera detection.

Experimental Setup

The system was rigorously tested in simulated real-world parking scenarios. Comprehensive test cases covered normal, peak, and edge conditions, with inputs like vehicle type and timestamps. Outputs included total duration, peak duration, surcharge, and transparent fare breakdown.

Sample Fare Calculation



Four-Wheeler Fare Breakdown

Base charge (3 hrs × ₹40): ₹120

Peak surcharge (2 hrs × ₹10): ₹20

Total: ₹140

The system automatically detects that 2 out of 3 hours overlapped with peak period, applying the surcharge only to those hours for fair billing.

Algorithm

- 1: Start program and initialize arrays, count = 0, nextToken = 1.
 - 2: Display main menu: Vehicle Entry, Vehicle Exit, View Active Vehicles, Exit
 - 3: Vehicle Entry: Check parking capacity, Generate unique token, Input number plate, Vehicle type, Entry time.
 - 4: Store details and mark vehicle as active.
 - 5: Vehicle Exit: Input number plate, Search matching active record, Input exit time, Convert time to minutes, Calculate total hours.
 - 6: Apply base rate: ₹25/hr (2W), ₹40/hr (4W)
 - 7: Calculate peak-hour overlap (6 PM–9 PM):Add ₹10/hr peak surcharge.
 - 8: Display total payable amount.
 - 9: Mark vehicle inactive.
 - 10: Repeat until user selects Exit.
 - 11: End program.
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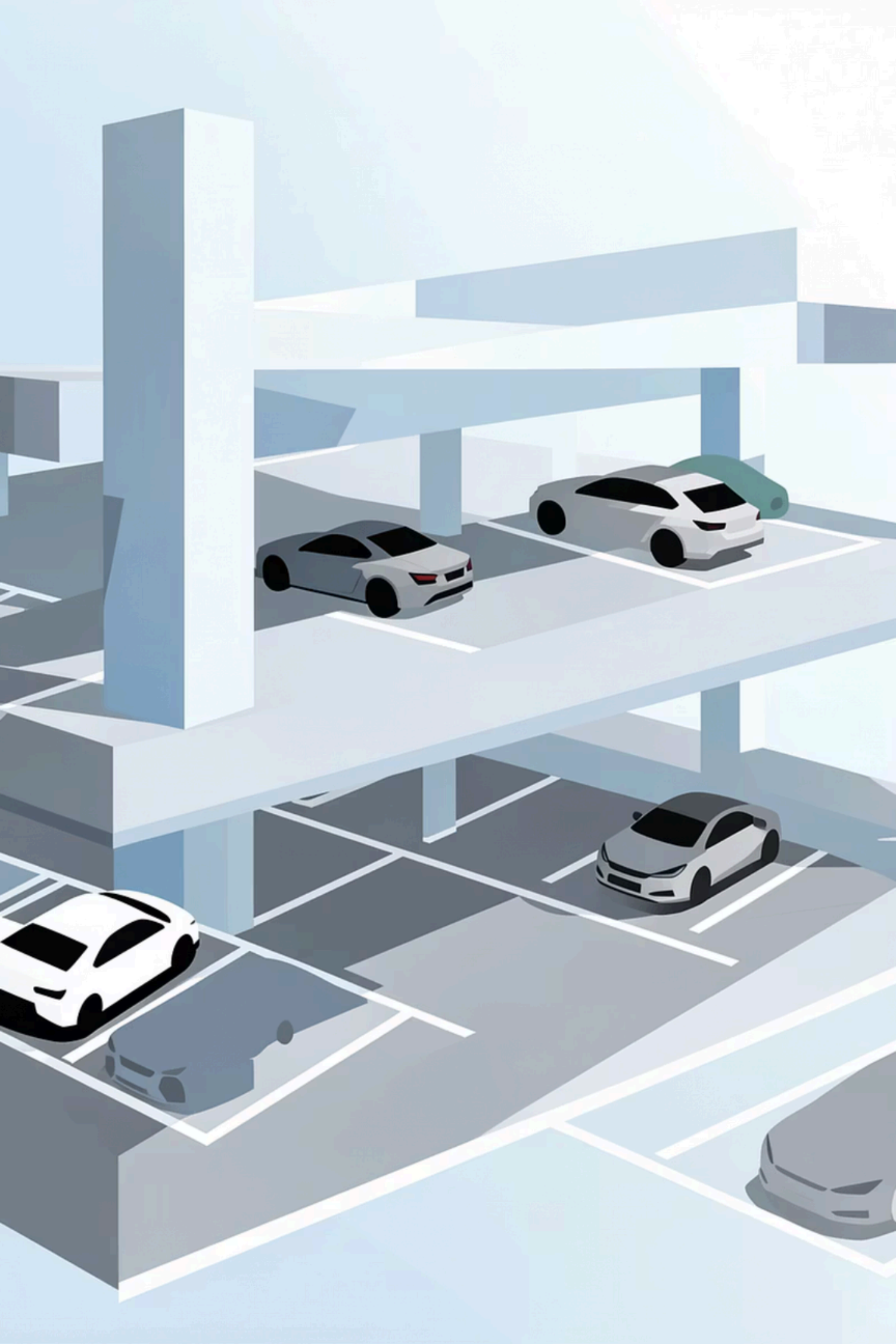
Results and Conclusion

Current Achievement

Successfully implemented an accurate, fair parking fee calculation system with transparent billing logic.

Future Enhancements

- Hardware camera integration for true automation
- Database storage for historical records and analytics
- Mobile payment gateway integration
- Real-time parking availability dashboard
- FASTag integration



Thank You